```
#include <stdio.h>
#define INFINITY 9999
#define MAX 10
void Dijkstra(int Graph[MAX][MAX], int n, int start);
void Dijkstra(int Graph[MAX][MAX], int n, int start) {
 int cost[MAX][MAX], distance[MAX], pred[MAX];
 int visited[MAX], count, mindistance, nextnode, i, j;
 // Creating cost matrix
 for (i = 0; i < n; i++)
 for (j = 0; j < n; j++)
 if (Graph[i][j] == 0)
 cost[i][j] = INFINITY;
 else
 cost[i][j] = Graph[i][j];
 for (i = 0; i < n; i++) {
 distance[i] = cost[start][i];
 pred[i] = start;
 visited[i] = 0;
 }
 distance[start] = 0;
 visited[start] = 1;
 count = 1;
 while (count < n - 1) {
 mindistance = INFINITY;
 for (i = 0; i < n; i++)
 if (distance[i] < mindistance && !visited[i]) {
 mindistance = distance[i];
 nextnode = i;
 }
 visited[nextnode] = 1;
 for (i = 0; i < n; i++)
 if (!visited[i])
 if (mindistance + cost[nextnode][i] < distance[i]) {
 distance[i] = mindistance + cost[nextnode][i];
 pred[i] = nextnode;
 }
 count++;
 }
 // Printing the distance
 for (i = 0; i < n; i++)
 if (i != start) {
 printf("\nDistance from source to %d: %d", i, distance[i]);
 } }
int main() {
```

```
int Graph[MAX][MAX], i, j, n, u;
n = 7;
Graph[0][0] = 0;
Graph[0][1] = 0;
Graph[0][2] = 1;
Graph[0][3] = 2;
Graph[0][4] = 0;
Graph[0][5] = 0;
Graph[0][6] = 0;
Graph[1][0] = 0;
Graph[1][1] = 0;
Graph[1][2] = 2;
Graph[1][3] = 0;
Graph[1][4] = 0;
Graph[1][5] = 3;
Graph[1][6] = 0;
Graph[2][0] = 1;
Graph[2][1] = 2;
Graph[2][2] = 0;
Graph[2][3] = 1;
Graph[2][4] = 3;
Graph[2][5] = 0;
Graph[2][6] = 0;
Graph[3][0] = 2;
Graph[3][1] = 0;
Graph[3][2] = 1;
Graph[3][3] = 0;
Graph[3][4] = 0;
Graph[3][5] = 0;
Graph[3][6] = 1;
Graph[4][0] = 0;
Graph[4][1] = 0;
Graph[4][2] = 3;
Graph[4][3] = 0;
Graph[4][4] = 0;
Graph[4][5] = 2;
Graph[4][6] = 0;
Graph[5][0] = 0;
Graph[5][1] = 3;
Graph[5][2] = 0;
Graph[5][3] = 0;
Graph[5][4] = 2;
Graph[5][5] = 0;
Graph[5][6] = 1;
```

```
 Graph[6][0] = 0; Graph[6][1] = 0;
 Graph[6][2] = 0;
 Graph[6][3] = 1;
 Graph[6][4] = 0;
 Graph[6][5] = 1;
 Graph[6][6] = 0;
 u = 0;
 Dijkstra(Graph, n, u);
 return 0;
}
```

Output:

```
  Distance from source to 1: 3
  Distance from source to 2: 1
  Distance from source to 3: 2
  Distance from source to 4: 4
  Distance from source to 5: 4
  Distance from source to 6: 3
```